
Autonomous Production Choke Controller

Final Submission Checklist

Honeywell Hackathon 2026
Engineering Submission

Document Reference: **HON-APCC-2026-001**
Revision: **2.0**
Date: **2026-07-25**
Classification: **Engineering Submission**

Official Honeywell Deliverables

Table 1: Official deliverable verification.

#	Deliverable	Status	Location
1	Python Source Code	COMPLETE	01_Source_Code/
2	Open-Loop Step Test Analysis	COMPLETE	05_Open_Loop_Step_Test_Analysis/
3	Dynamic Model Identification	COMPLETE	06_Dynamic_Model_Identification/
4	Controller Design	COMPLETE	07_Controller_Design/
5	Scenario A Results	COMPLETE	09_Scenario_Results/
6	Scenario B Results	COMPLETE	09_Scenario_Results/
7	Scenario C Results	COMPLETE	09_Scenario_Results/
8	Target Oil Rate Plots	COMPLETE	12_Generated_Plots/

9	Actual Oil Rate Plots	COMPLETE	12_Generated_Plots/
10	Wellhead Pressure Plots	COMPLETE	12_Generated_Plots/
11	Flowline Pressure Plots	COMPLETE	12_Generated_Plots/
12	Bottom Hole Pressure Plots	COMPLETE	12_Generated_Plots/
13	Choke Position Plots	COMPLETE	12_Generated_Plots/
14	Technical Report	COMPLETE	02_Technical_Report/
15	Presentation	COMPLETE	03_Presentation/
16	Streamlit Dashboard	COMPLETE	04_Streamlit_Dashboard/
17	README	COMPLETE	14_Project_Documentation/
18	requirements.txt	COMPLETE	01_Source_Code/
19	Validation Report	COMPLETE	08_Validation_Report/
20	Architecture Diagrams	COMPLETE	10_Architecture_Diagrams/

Supporting Deliverables

Table 2: Supporting deliverable verification.

#	Deliverable	Status	Location
21	Workflow Diagrams	COMPLETE	11_Workflow_Diagrams/
22	Reference Dataset Comparison	COMPLETE	13_Reference_Dataset_Comparison/
23	Execution Guide	COMPLETE	15_Execution_Guide/
24	Project Documentation	COMPLETE	14_Project_Documentation/
25	Example Outputs	COMPLETE	16_Example_Outputs/
26	Submission Checklist	COMPLETE	17_Final_Submission_Checklist/
27	Submission Package	COMPLETE	18_Submission_Ready_ZIP/

Compliance Verification

Table 3: Honeywell requirement compliance.

Requirement	Status
Simulator-generated data for system identification	COMPLIANT
FOPDT models from step test experiments only	COMPLIANT
Reference dataset NOT used for identification/tuning/design	COMPLIANT
Reference dataset is post-hoc consistency check only	COMPLIANT
Defense-in-depth safety with 4 layers	COMPLIANT
Constraint enforcement on WHP, FLP, BHP	COMPLIANT
Ramp rate limits respected	COMPLIANT
All 3 validation scenarios executed	COMPLIANT
Reproducible (seed=42)	COMPLIANT

Final Statement

ALL DELIVERABLES COMPLETE. READY FOR SUBMISSION.